

What the Instrument Can See

Measuring the coverage boundary of a social-media cybersecurity sensor

ccint — iteration report

22 September 2026

In one paragraph. The previous phase of this project established, by permutation test, that no topic trend in the corpus is distinguishable from noise, and measured that roughly eight times the data would be needed before one could be. That leaves an obvious question: is the absence in the world or in the instrument? This iteration built the measurements needed to answer it, and the answer is specific. For the one event class we can verify against an external registry — Canadian ransomware victims — the platform carries the events almost perfectly (26 of 27) and the collector captures almost all of what it carries (26 of 27), but essentially none of it is human discussion: 144 relevant posts from 15 accounts, of which three accounts and five posts are not automated feeds, and all five concern a single event. What looked like coverage is syndication. The corpus is a near-complete mirror of a handful of threat-intelligence bots and a near-empty record of Canadian public attention to cyber incidents.

1 The question this iteration answers

The system collects Canada-related cybersecurity discussion from Bluesky, labels it under versioned schemes, and looks for temporal change. Its headline result so far is negative and well-characterised: the largest topic movement in the 30-day window, `data_breach` at +22%, has $p_{\text{FWER}} = 0.734$ under an author-level circular-shift null, and the measured power requirement is about $8\times$ the current volume.

A negative result of that kind is only interesting if you know what produced it. Three explanations were open:

1. **The world.** Canadian cyber discussion genuinely has no structure at this timescale.
2. **The gate.** The relevance judge is too strict, so the corpus is a biased subsample of what was collected.
3. **The boundary.** The collector never retrieved the relevant material in the first place, so no judge could recover it.

Nothing in the system could separate these, because every number it produced came from labels it had produced itself. This iteration added the three measurements that can, and they resolve in an order that was not the expected one.

2 What was measured

2.1 The collection boundary had never been examined

The collector issues one search per term over a fixed list of 34 cybersecurity terms — 25 English, 9 French — and merges the results. None of the 34 mentions Canada. The list has not changed since the first day of the project; the `query_version` mechanism faithfully recorded that it was `cyber_v1`, which guarantees reproducibility but says nothing about adequacy.

Because Bluesky search is close to text matching, per-term productivity can be reconstructed offline from the 91,551 posts already stored, without new collection. Figure 1 plots what each term cost against what it returned.

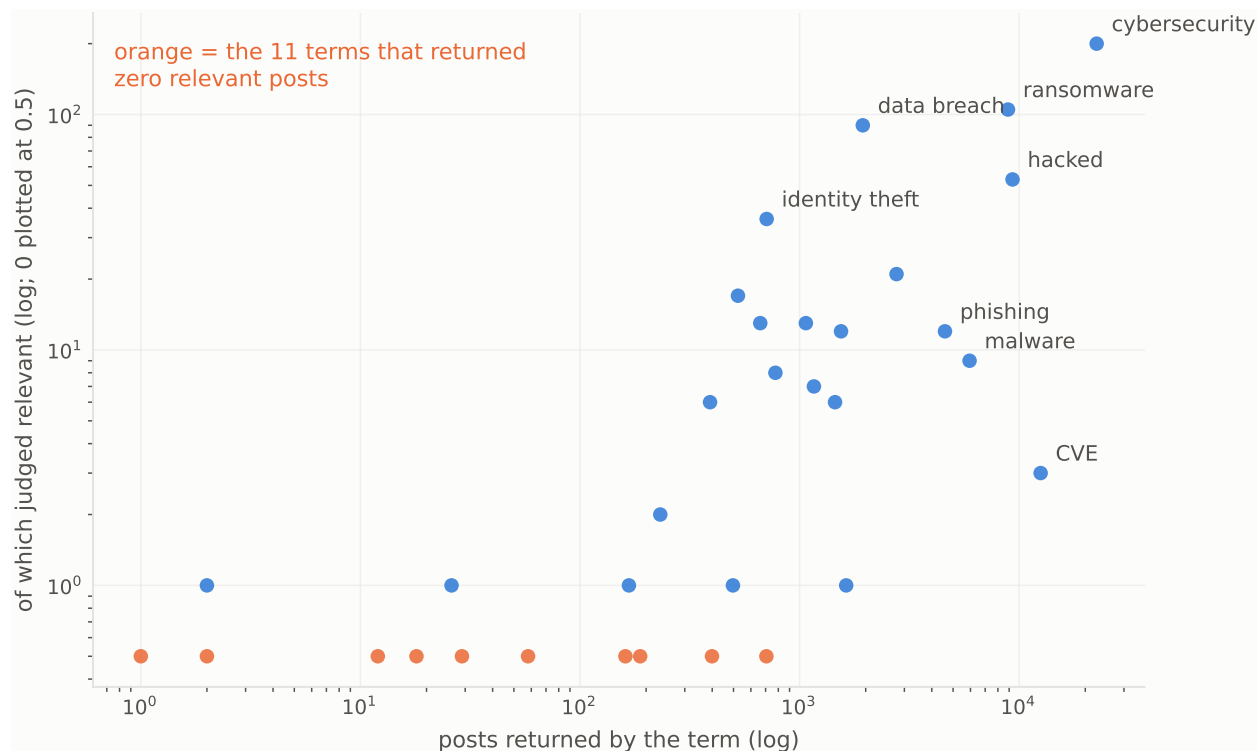


Figure 1: Each search term: posts retrieved against posts subsequently judged relevant. Both axes are logarithmic; terms with zero relevant posts are drawn at 0.5 so they remain visible. The distribution is not merely uneven — it is close to degenerate.

Three properties matter more than the ranking:

- CVE retrieved 12,502 posts and yielded 3 relevant ones, a precision of 0.024%. It accounts for 13.7% of everything collected.
- Eleven terms returned zero relevant posts. One of them, `maliciel`, returned no posts at all in 30 days.
- `cybersecurity` alone accounts for 30.8% of all relevant posts and 24.6% of the entire corpus. The time series of the corpus is, to first order, the time series of one search term. Nothing monitors it.

The instructive part is the provenance. The project’s specification named four term pairs and wrote “etc.”; the remaining 26 terms were supplied during implementation from general domain knowledge. They were never derived from data, never compared against an external list, and never revised. The published work that does this well does one of two things: it inherits a list from practitioners — Tweezers (NDSS 2025) takes its keywords from a threat intelligence company’s analysts — or it grows one from a seed by measured expansion. Tweezers also reports what such lists cost: comparing its own against the earlier W2E list, more than half of the tweets it collected would have been missed by W2E, and W2E was itself missing `phishing`,

cve and patch. Two lists built by professionals disagreed by more than half.

2.2 The relevance gate, replaced and then measured

The old gate was `is_relevant = has_cyber_term AND has_canada_term AND corroborated`. Both sides are narrow instruments joined by a conjunction, so recall is bounded by their intersection: of 91,551 posts collected, 650 passed (0.72%), and the 89,714 rejected had never been examined by anything semantic.

The replacement asks a model two questions, and the first design decision was to keep them apart. A single prompt asking whether a post is about *cybersecurity in Canada* produced a 6.8% pass rate on rejected posts, and inspection showed why: the model had collapsed the conjunction onto the side it knows better. The passing posts were Fortinet advisories, Lynx victim listings, and a French story about the French Ministry of Education. Asked as two independent questions, the Canadian side passed 1.08%, and the passing posts began to include material that should have been there all along — a Quebec cabinet minister responsible for cybersecurity, .ca law-firm domains, “GC website” as colloquial shorthand for the federal government.

A second decision follows from the corpus rather than from the model: ask about Canada first, over everything, and ask about cyber only where a Canadian entity was found. Every post in this corpus arrived through a cybersecurity search term, so the cyber side is nearly always true and carries little information; the Canadian side is the rare and binding one. Running it this way also produces an entity map of the whole corpus as a by-product, which is the input to query expansion and to event linking, and it reduces the call count from $2N$ to $N + \varepsilon N$.

The gate’s own failure, and its ceiling. The first version of the entity prompt recovered Canadian entities in only 55.4% of registry-verified positive posts — and, more diagnostically, in only 69.0% of posts that contain the literal string *Canada* or *Canadian*. Specificity was excellent at 99.5%. The failure was therefore recall, and its cause was the prompt: a clause instructing the model that an empty list is the expected answer, followed by a long list of prohibitions, drove a 4B model to the null answer. Removing the restraint clause and adding an explicit rule — if the country name appears literally, it counts — moved literal-mention recall from 69.0% to 99.3% (Figure ??).

And the fix broke something the evaluation could not see. The corrected prompt raised corpus relevance from 906 posts to 3,528. That number is wrong. Inspection of the additions found *trustedsec.com*, *malwarebytes.com*, the UK’s NCSC, Revolut, a French story about the *Éducation nationale*, and *ia.cr* — the Costa Rican top-level domain. The prompt’s clause “*a .ca or .gc.ca domain*” had been generalised by the model into “*a domain*”.

Two mechanical checks, requiring no further model calls, quantify it:

Extracted entity type	Count	Not present in the post text
place	2,192	1,053 (48.0%)
org	1,364	12 (0.9%)
domain	1,296	49 (3.8%)
gov	889	10 (1.1%)

Of the 1,296 domain entities, 312 (24.1%) actually end in a Canadian TLD.

A verification channel obtained by accident. Character offsets are computed with `str.find` rather than requested from the model, on the narrow grounds that small models miscount offsets. That decision turned into an independent hallucination detector: an entity string that cannot be located in the post was invented. Nearly half of all `place` entities were — the model writing “Canada” into a French post about a French ministry. Declining to trust a model with a task it does badly produced a free check on the task it was actually given.

Applying both rules — discard entities absent from the text, discard domains outside Canadian TLDs — removes 2,086 entities and 1,180 posts, and leaves the picture below. The validation layer is retained in the labeler even though the prompt will be corrected, because it is deterministic, costs nothing, and catches exactly the class of error the external evaluation set cannot.

Relevance gate	recall @ registry pos.	specificity @ registry neg.	corpus relevant
rules_v2 — keyword conjunction	46.5%	99.8%	650
llm_v2a — two questions, prompt v1	53.5%	99.7%	860
llm_v2c — prompt v2 + validation	68.3%	99.4%	2,348

before validation, prompt v2 reported 69.3% / 99.0% / 3,528 — one third of that corpus was spurious.

The negative set certified a corpus that was a third wrong. A specificity of 99.0% on `E_reg_neg` implies roughly 890 false positives across 89,044 posts. The true figure was 1,181. The negative set did not detect the failure because it is composed of feed posts about foreign ransomware victims: formulaic English, few URLs, no French. The model’s actual failure modes — over-generalising a domain rule, hallucinating a country name into French text — do not occur in that population. An external label removes self-reference; it does not confer representativeness, and a non-representative evaluation set does not merely fail to detect a regression, it actively certifies one.

What remains is a knowledge ceiling. Of the posts the corrected and validated extractor still misses, **none** contains the word *Canada* in any form. They read “*New post from #Qilin: Coldfish Seafood*” — a real Canadian victim, named, with no country marker anywhere in the text. No amount of prompting will let a 4B model know that Coldfish Seafood is Canadian.

This is a ceiling of a particular kind. The residual error in the relevance gate is not a model-capability problem and not a prompt problem: it is missing world knowledge about small Canadian organisations. The only thing that resolves it is a gazetteer — and the gazetteer that would resolve it is the external registry currently being used to measure the gap. The measurement instrument and the repair are the same object.

2.3 An external reference, and what it revealed about itself

Until this iteration every precision and recall figure in the project traced back to a single author: the person who wrote the rules also wrote the prompts and produced the hand-coded evaluation set. A reported $\kappa = 0.870$ measures agreement under those conditions, not independence.

The registry breaks that loop. Victim records were ingested from ransomware.live for Canada (positives) and the United States, France and the United Kingdom (negative entity sources): 1,080 events and 1,943 entities. Matching victim organisation names against the corpus produces evaluation sets whose labels no part of this system produced.

Two constraints were imposed on their construction, and both turned out to matter more than expected.

First, the definition of an evaluation set may not contain the output of the system being evaluated. A natural specification for the negative set was “mentions a foreign victim, and `rules_v2` judged `is_canada` false”. That specification makes `rules_v2` precision identically 1 on the set, and any replacement can only appear worse — an artefact of construction with no relation to capability. The negative set uses registry conditions only.

Second, the positive set must be stratified by account type. The reason is visible in the data: registry victim names appear on Bluesky almost exclusively inside the accounts that syndicate the registry. Of 101 positive posts, 100 come from automated feeds; the three largest contributors are `ransomlook`, `cyberintelligence` and `ransomware.live` itself. Median engagement is zero and 94.9% of the posts have no likes at all. The single post classified as non-feed turned out to be a golf-industry content farm auto-generating an article about a golf club’s ransomware incident, with zero likes, zero reposts and zero replies.

A metric that rewards capturing bots. Had `recall@E_reg_pos` been adopted as the optimisation target, it would have climbed towards 100%, because those posts are formulaic and a model need only learn that “Inglewood Golf” is a Canadian company. The metric would improve substantially while the research question — what Canadians say about cyber incidents — received no answer at all. External labels are necessary but not sufficient: an external label attached to a non-independent *population* reintroduces the circularity it was meant to remove.

3 The central result

Combining the registry with a targeted per-event search over the platform separates the three explanations. For every Canadian event in the window, the victim organisation’s name was searched directly on Bluesky — with no cybersecurity term attached — over a $[-2, +30]$ day window. Because bare organisation names attract unrelated traffic (*MEQ* retrieves the medical unit; *Air Canada* retrieves fare deals), every hit was then passed through the cyber judgement before counting. This is entity-directed retrieval: it establishes an upper bound on what the platform could have yielded for an event whose identity is already known.

Stage	Events	Share
Listed in the registry, in window	27	—
At least one cyber-related post exists on Bluesky	26	96%
At least one such post was collected by ccint	26	96%
At least one such post came from a non-feed account	2	7%

The 144 cyber-related posts across these events come from 15 accounts. Twelve are automated threat-intelligence feeds. The three that are not are Canadian news organisations — `ctvnewsmontreal`, `mediasca`, `985fm.ca` — contributing five posts between them, all about the same event. Individual users contributed nothing.

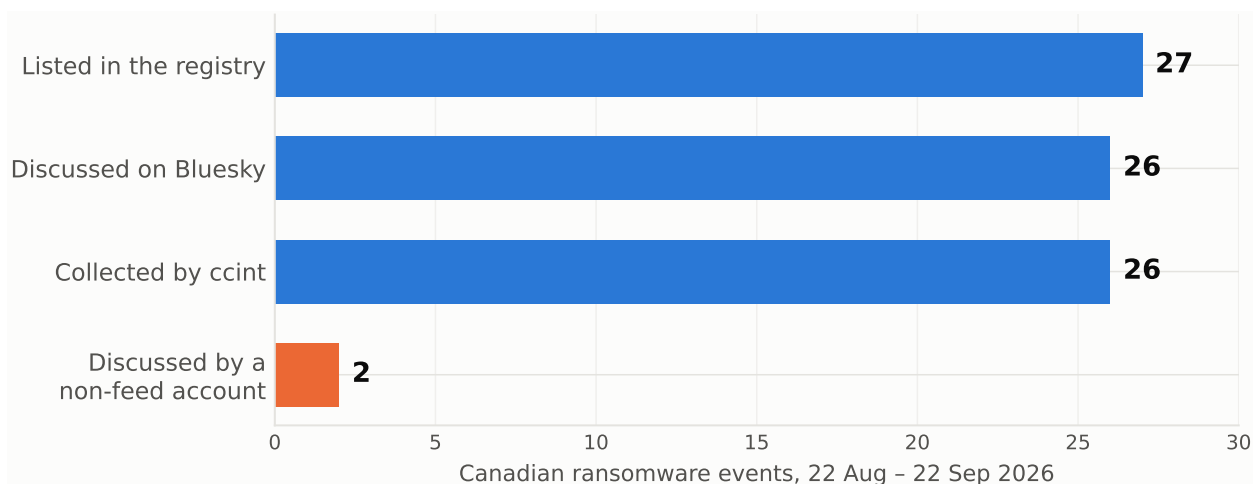


Figure 2: The same 27 Canadian ransomware events at four successive gates. The attrition is almost entirely at the last one, and the last one is not a property of this system.

The collector is not the bottleneck for this event class. It captured 26 of the 26 events the platform carried. The reason the corpus contains no detectable trend in ransomware discussion is that there is almost no ransomware *discussion* to detect — there is a nearly complete mirror of several leak-site feeds, and the corpus reproduces it faithfully.

3.1 The one event that behaved differently

The exception is worth stating in full, because it is the only case in the month where the platform behaved the way the project assumed it would.

The Commission de la construction du Québec, a provincial construction industry regulator, was listed by the Qilin ransomware group; roughly 350,000 people’s data was taken. This event drew genuine Canadian media coverage on Bluesky, in French and English: *ctvnewsmontreal* on a class action filed in Quebec Superior Court, *mediasca* on Russian attackers taking the records of 350,000 people, *lapresse.ca*, and *985fm.ca* carrying an interview with the CCQ’s chief executive.

Four of those five posts are in the corpus. The one that is not reads “*Une protection préventive mise en place à la CCQ: «Je me sens responsable »*”. It contains no term from the 34-term list — *protection préventive* is not in any lexicon — and so it was unreachable. This is the collection-boundary gap at human scale: one post, one missing phrase, a real event, a real Canadian outlet.

3.2 Two gaps, and a reference that can only see one

The measurements point at two distinct deficiencies, and conflating them would be a mistake.

	Registry-verifiable events	General Canadian cyber talk
Collection recall	26/27 events (96%)	unmeasured; at least 901 rejected posts carry a Canadian entity
Binding constraint	the platform has no organic discussion	the Canada lexicon misses entities
Fixable by better queries?	no	yes
Visible to the registry?	yes	no

A full-corpus screen of the 83,370 rejected posts long enough to judge found 901 containing a Canadian entity; attribution shows 55.8% failed because `is_canada` was false — the lexicon missed the entity — and *none* were blocked by the corroboration rule that a prior review had proposed retiring. Replacing the gate raised corpus-level relevant posts from 650 to 2,348. Of the 1,799 posts newly admitted, 84.0% were rejected by the Canadian side of the conjunction — scored `is_cyber` true and `is_canada` false — against 12.2% on the cyber side and 3.7% on both. In the other direction the new gate rejects 138 posts the rules had admitted. The asymmetry is the point: nearly all recoverable recall sat behind the Canada lexicon, and none of it behind the corroboration rule that a prior review had proposed retiring.

But the registry cannot see any of that population. It contains ransomware victims, so it can only score the class of discussion that ransomware feeds dominate. An external reference removes self-reference; it does not by itself deliver representativeness.

4 What this says about what the system is

The project was framed as a longitudinal study of how Canadian cybersecurity discussion changes. The measurements above make a weaker claim available and a stronger one unavailable.

Unavailable: trend analysis of Canadian cyber discourse on this platform, at this volume, for this event class. Not because the statistics are under-powered — though they are — but because the underlying signal is mostly machine output. Adding data would add feed posts. The $8\times$ figure from the power analysis is an honest answer to the wrong question; $8\times$ of this corpus is $8\times$ as much syndication.

Available, and unusual: a sensor that reports its own blind spots with numbers. Nobody in this literature publishes a denominator. Tweezers reports detecting 84 of 151 events; systems it compares against report 45 and 4. All three report a numerator. None reports how many events had no platform presence at all, because establishing that requires the per-event entity-directed retrieval run here, and that is only cheap if the event count is small. This project's greatest liability — a corpus of 650 relevant posts — is exactly what makes the denominator measurable.

The defensible contribution is not “we detected Canadian cyber events on Bluesky”. It is “here is what a single-platform social sensor can and cannot see about a national cybersecurity picture, measured against an external registry, with the syndication component separated from the organic one”. The first claim is weak and partly false. The second is, as far as the surveyed literature goes, unreported.

5 Engineering changes supporting the above

Change	Why it was needed
Single taxonomy source	The topic list existed twice: a YAML file for the rules labeler and a Python tuple for the model's decoding enum. Merging them surfaced a live drift — 10 buckets versus 11 — so the two label versions had been classifying into label spaces of different size, and every diff between them carried a component unrelated to model behaviour.
Prompt provenance	Prompts were Python string literals with no record anywhere. A model labeler's output is entirely determined by its prompt, so "labels are never overwritten" preserved the labels while losing the conditions that produced them. Prompts are now versioned files; the hash of the body goes to a side table, which keeps the no-update rule free of exceptions.
Three-state <code>is_cyber</code>	The gate judges cyber only where a Canadian entity was found, and "never asked" was being stored as "asked and answered no" for 87,813 rows. This is the same error the project had already ruled out in its actor typology, where <code>unclassified</code> must never be read as negative.
Registry tables	<code>external_events</code> , <code>external_event_entities</code> , <code>eval_sets</code> , <code>post_entities</code> . Evaluation sets are frozen on creation and carry a stratum.
Measurement-only retrieval mode	An entity-directed retrieval path still writes to <code>collection_runs</code> ; the schema constraint previously forced such paths to bypass run bookkeeping, which is precisely what makes system failures indistinguishable from social change.

6 Limitations

- **One registry, one event class.** `ransomware.live` lists only victims named on leak sites. Breaches disclosed by regulators, phishing campaigns, advisories and policy events are entirely invisible to it, as are victims who paid. Every figure conditioned on the registry describes ransomware coverage, not Canadian cybersecurity coverage.
- $N = 27$. The three-stage decomposition rests on 27 events in one 30-day window. The qualitative pattern is stark, but the percentages carry wide intervals and should not be quoted as point estimates.
- **Feed classification is a heuristic.** Accounts are labelled automated by volume and near-zero engagement. The golf content farm shows the rule's failure mode: a low-volume automated account is scored as organic, which biases the organic count *upward*. The true organic count is therefore no higher than reported, and probably lower.
- **No independent human annotation yet.** A 200-post stratified audit sample has been drawn and deliberately left unlabelled. The person who wrote the prompts should not be the person who scores them; doing so would reproduce the circularity that the registry was introduced to break.
- **The evaluation sets are narrow in a way that matters.** Both are built from ransomware victim names, so they exercise a formulaic English subpopulation. The domain over-generalisation documented above passed them cleanly. Any future prompt change should be checked against a probe drawn from the corpus itself, not only against the registry sets.

- **A 4B model.** Both gates run on Qwen3-4B. The residual failure mode identified above — not knowing which small firms are Canadian — would partly yield to a larger model, though a gazetteer addresses it more directly and more cheaply.

7 What follows

1. **Widen the registry before widening anything else.** Advisories from the Canadian Centre for Cyber Security, CISA KEV and Canadian media RSS would extend the external reference past ransomware, which is the single change that would most alter the picture above. The current denominator is narrow in a way that flatters nothing.
2. **Feed the registry back into the gate.** The irreducible residual is missing knowledge of Canadian organisation names, and the registry is a list of exactly those names. This closes a loop that is currently open in the wrong direction.
3. **Add a Canadian query stream.** Not to improve ransomware coverage, which is already at 96%, but to reach the population the registry cannot see — the CCQ interview that no cybersecurity term could retrieve.
4. **Have someone else label the audit sample.** It is the only remaining step that cannot be automated, and the one every other number depends on.
5. **Report absolute counts, never recall ratios.** With a denominator this small and this biased, a ratio invites a comparison that the data cannot support.

All figures in this report derive from the corpus of 91,551 posts collected between 22 August and 22 September 2026, the ransomware.live registry as retrieved on 22 September 2026, and an entity-directed retrieval pass executed the same day. Labels are stored under versioned identifiers and every model-produced label carries the hash of the prompt that produced it. The test suite stands at 241 passing.